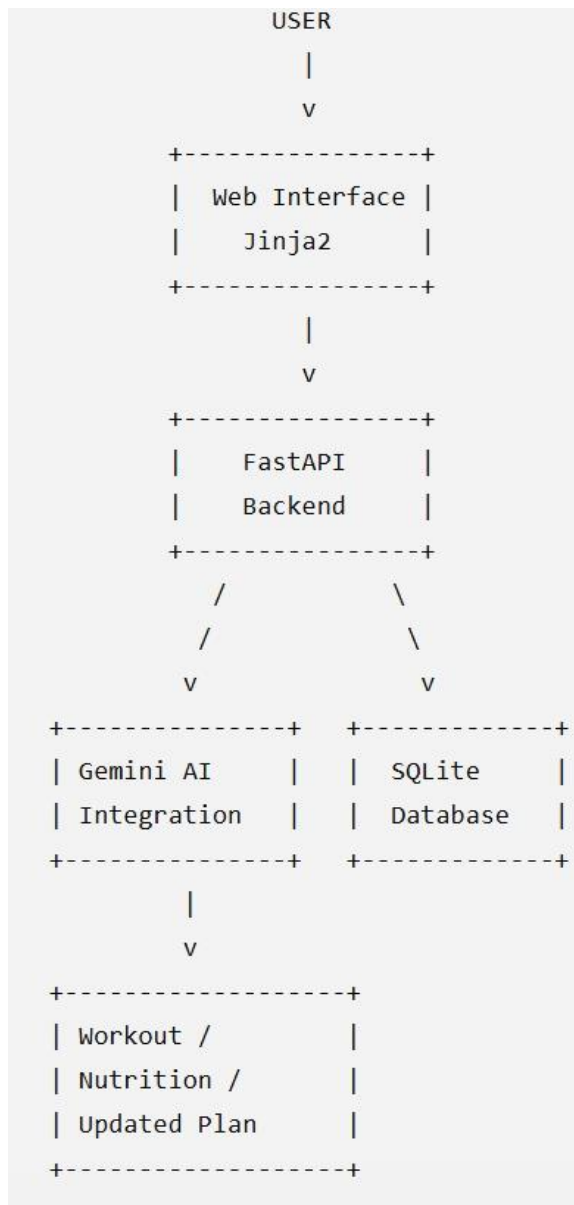


3.PROJECT DESIGN PHASE

The Project Design Phase converts the requirements into a technical architecture. FitBuddy follows a modular web application architecture consisting of the frontend, backend, AI services, database, and administrative interface.

3.1 System Architecture

The basic architecture of FitBuddy can be represented as:



3.2 Frontend Design

The frontend is designed using HTML and CSS with Jinja2 templates. The main pages include:

- Home page.
- Workout result page.
- Feedback section.
- Administrator dashboard.

The home page collects user information. The result page displays the generated seven-day workout plan and nutrition/recovery information.

3.3 Backend Design

FastAPI is used as the backend framework. It manages:

- HTTP requests.
- Form processing.
- Data validation.
- Gemini communication.
- Database operations.
- API endpoints.
- Administrative authentication.

3.4 AI Design

The AI component is divided into separate services.

The main Gemini planner generates the seven-day workout plan.

A separate Gemini service generates nutrition and recovery guidance.

Another AI service processes user feedback and produces an updated workout plan.

This modular approach makes the application easier to maintain and extend.

3.5 Database Design

SQLite is used as the database because it is lightweight and suitable for the project.

The main database entities are:

User

```
User
-----
id
user_id
username
age
weight
goal
intensity
created_at
```

Plan

```
Plan
-----
id
user_id
original_plan
updated_plan
nutrition_tip
feedback
created_at
updated_at
```

The database design allows original and updated plans to be retained.

